A Closed-Form Gaussian-Mixture Toy Model for Validating the Girsanov-Corrected SMC Weight

Validation note accompanying the main derivation

Abstract

This note constructs a one-dimensional Gaussian-mixture toy problem in which the prior, the noised marginals, the exact score and denoiser, the true posterior, and – crucially – the *exact* intermediate likelihood $p(y \mid x_\sigma)$ are all available in closed form. Because the exact intermediate likelihood is normally intractable, this toy model lets us simulate both the approximate-guided diffusion actually used in practice and an exact-guided diffusion that targets the true posterior exactly, and to check the Girsanov/Doob-transform SMC correction against a genuine analytical ground truth rather than against itself. We derive the relevant quantities from scratch, discuss a sign-convention pitfall that this exercise uncovered in the drift of the reverse-time SDE, and report numerical results: the corrected sampler converges toward the analytical posterior at the canonical Monte Carlo rate in the number of particles, up to a small residual bias tied to the time-discretization step count.

1 The toy model

We work in one dimension with a K-component Gaussian-mixture prior,

$$p(x_0) = \sum_{k=1}^{K} w_k \, \mathcal{N}(x_0; \mu_k, \sigma_k^2), \tag{1.1}$$

noised by the same variance-exploding (VE), EDM-style forward process used throughout the main derivation [3]:

$$x_\sigma = x_0 + \sigma \, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, 1), \tag{1.2}$$

with noise level identified with physical time, $\sigma(t) = t$. We use $K = 2$, $w = (0.5, 0.5)$, $\mu = (-3, 3)$, $\sigma_k^2 = 1$ throughout the numerical experiments below.

Remark 1.1. *The conditional law $x_\sigma \mid x_0 \sim \mathcal{N}(x_0, \sigma^2)$ is exactly Gaussian at* every *finite $\sigma > 0$ – there is nothing asymptotic about it. The marginal $p_\sigma(x_\sigma) = \int p(x_\sigma \mid x_0) p(x_0)\, dx_0$ is a Gaussian mixture (below) at every finite σ as well; it only becomes* practically *indistinguishable from a single Gaussian as $\sigma \to \infty$, once the fixed inter-component separation becomes negligible relative to the growing noise scale.*

Convolving a Gaussian mixture with independent Gaussian noise leaves the weights and means untouched and inflates each component's variance:

$$p_\sigma(x) = \sum_k w_k \, \mathcal{N}(x; \mu_k, \sigma_k^2 + \sigma^2). \tag{1.3}$$

We adopt a K-component Gaussian, linear observation model,

$$y = x_0 + \eta, \qquad \eta \sim \mathcal{N}(0, r^2). \tag{1.4}$$

A single Gaussian prior would make the Bayes-optimal denoiser affine in x, and the plug-in approximate likelihood introduced below would then be proportional to the true intermediate likelihood – same mode, only a scale difference – too easy a test of the correction. The mixture prior makes the denoiser genuinely nonlinear, so the approximation gap has the same qualitative character it has with a real trained network.

2 Exact quantities

Write $V_k(\sigma) := \sigma_k^2 + \sigma^2$ and let

$$\gamma_k(x, \sigma) := \frac{w_k \, \mathcal{N}(x; \mu_k, V_k(\sigma))}{\sum_j w_j \, \mathcal{N}(x; \mu_j, V_j(\sigma))} \tag{2.1}$$

be the posterior mixture responsibilities of the noised marginal (1.3). Differentiating $\log p_\sigma$ component-wise gives the exact score,

$$\nabla_x \log p_\sigma(x) = \sum_k \gamma_k(x, \sigma) \left(-\frac{x - \mu_k}{V_k(\sigma)} \right), \tag{2.2}$$

and the exact Bayes-optimal denoiser $D(x, \sigma) := \mathbb{E}[x_0 \mid x_\sigma = x]$,

$$D(x, \sigma) = \sum_k \gamma_k(x, \sigma) \left[\mu_k + \frac{\sigma_k^2}{V_k(\sigma)} (x - \mu_k) \right], \tag{2.3}$$

consistent with Tweedie's identity $D(x, \sigma) = x + \sigma^2 \nabla_x \log p_\sigma(x)$ [2]. Neither (2.2) nor (2.3) requires a trained network – both replace D_θ exactly, isolating the Girsanov/SMC machinery from any network-approximation error.

The posterior $p(x_0 \mid y) \propto p(y \mid x_0) p(x_0)$ under (1.4) is again a mixture, with the standard per-component conjugate update; this is our ground truth.

The intermediate likelihood $p(y \mid x_\sigma)$ – normally intractable, and the reason approximate guidance is used in practice at all – is available here in closed form. Writing the per-component denoising posterior $x_0 \mid x_\sigma = x, \, k \sim \mathcal{N}(m_k(x, \sigma), c_k(\sigma))$ with

$$m_k(x, \sigma) = \mu_k + \frac{\sigma_k^2}{V_k(\sigma)} (x - \mu_k), \qquad c_k(\sigma) = \frac{\sigma_k^2 \sigma^2}{V_k(\sigma)}, \tag{2.4}$$

and convolving with the observation noise gives

$$p(y \mid x_\sigma = x) = \sum_k \gamma_k(x, \sigma) \, \mathcal{N}\big(y; \, m_k(x, \sigma), \, r^2 + c_k(\sigma)\big). \tag{2.5}$$

The approximate (plug-in) likelihood actually used for guidance in practice replaces the mixture (2.5) with a single Gaussian at the denoised point estimate,

$$\tilde{p}_\theta(y \mid x_\sigma = x) = \mathcal{N}\big(y; \, D(x, \sigma), \, r^2\big). \tag{2.6}$$

Two gaps between (2.5) and (2.6) are visible directly from the formulas: the exact version is a genuine mixture over the per-component means m_k, which the plug-in approximation collapses to a single point; and the exact version's variance is inflated by $c_k(\sigma)$ – the residual uncertainty about x_0 given only the noisy x_σ – which the plug-in approximation ignores entirely, silently treating the denoiser's point estimate as if it were the true x_0. That second gap is exactly what the Girsanov correction exists to fix.

3 Exact and approximate guided diffusion

Let $\tau = T - t$ be the sampling-time reparametrization ($\tau = 0$ at pure noise, $\tau = T$ at clean data) and $\bar{a}(\tau) := a(T - \tau)$ with $a(t) := d\sigma(t)^2/dt = 2\sigma(t)$ under $\sigma(t) = t$. Anderson's reverse-time SDE theorem [1, 5] gives the unconditional reverse process directly in terms of the true score, unambiguously:

$$dZ_\tau = \bar{a}(\tau) \, \nabla_x \log p_{\sigma(\tau)}(Z_\tau) \, d\tau + \sqrt{\bar{a}(\tau)} \, dW_\tau, \qquad \sigma(\tau) := \sigma_{T-\tau}. \tag{3.1}$$

Adding the gradient of a log-likelihood ℓ_τ to the drift steers the process toward the corresponding posterior:

$$dZ^\bullet_\tau = \bar{a}(\tau)\Big[\nabla_x \log p_{\sigma(\tau)}(Z^\bullet_\tau) + \nabla_x \ell_\tau(Z^\bullet_\tau)\Big]\, d\tau + \sqrt{\bar{a}(\tau)}\, dW_\tau. \tag{3.2}$$

Taking $\ell_\tau(x) := \log p(y \mid x_{\sigma(\tau)} = x)$, the *exact* intermediate likelihood (2.5), gives the exact-guided diffusion, whose $\tau = T$ marginal is the true posterior exactly. Taking $\ell_\tau(x) := \log \tilde{p}_\theta(y \mid x_{\sigma(\tau)} = x)$, the plug-in likelihood (2.6), gives the approximate-guided diffusion actually simulated by a practical guided sampler – and whose $\tau = T$ marginal is biased, for the reasons noted above.

Remark 3.1 (A sign-convention pitfall)**.** *It is common in the EDM/denoiser literature [3] to write $s_\theta(x, \sigma) := (x - D(x, \sigma))/\sigma^2$, chosen so that the deterministic probability-flow ODE step comes out in the clean form $(x - D)/\sigma$. By Tweedie's identity this s_θ is the* negative *of the true score, $s_\theta = -\nabla_x \log p_\sigma$. Reverse-SDE formulas of the form $dX_t = -a(t)s_\theta(X_t, t)\, dt + \ldots$, standard throughout the score-based generative modeling literature, are only correct as written when s_θ denotes the true score directly; substituting the EDM-convention s_θ into such a formula introduces a spurious overall sign error in the unconditional part of the drift. We found exactly this: writing the drift of (3.1) with $+s_\theta$ in place of $\nabla \log p_\sigma$ causes the unguided reverse SDE to diverge (verified numerically – see §5), even though the formula is a one-symbol substitution away from (3.1) and each convention is individually standard in its own home literature. Writing the drift directly in terms of $\nabla_x \log p_\sigma$ throughout, as in (3.1)–(3.2), avoids the ambiguity entirely and is what we implement.*

Both SDEs are simulated by Euler–Maruyama on a decreasing σ grid (physical time, matching a practical EDM sampler rather than the increasing-τ convention used above for the derivation), with step $\Delta = \sigma_{\text{cur}} - \sigma_{\text{next}} > 0$:

$$x_{\text{next}} = x_{\text{cur}} + \bar{a}(\sigma_{\text{cur}})\Big[\nabla \log p_{\sigma_{\text{cur}}}(x_{\text{cur}}) + \nabla \ell(x_{\text{cur}}, \sigma_{\text{cur}})\Big]\Delta + \sqrt{\bar{a}(\sigma_{\text{cur}})\,\Delta}\;\xi, \qquad \xi \sim \mathcal{N}(0,1). \tag{3.3}$$

4 The Girsanov/Doob-transform correction

Let P^{pr} and P^{ap} denote the path measures of (3.1) and of (3.2) with the *approximate* likelihood, respectively. Itô's formula applied to $\ell_\tau(Z^{\text{ap}}_\tau)$ under P^{ap} gives

$$d\ell_\tau(Z^{\text{ap}}_\tau) = \Big[\partial_\tau \ell_\tau + b^{\text{ap}} \cdot \nabla \ell_\tau + \tfrac{1}{2}\bar{a}\, \Delta \ell_\tau\Big] d\tau + \sqrt{\bar{a}}\, \nabla \ell_\tau^\top\, dW^{\text{ap}}_\tau, \tag{4.1}$$

with $b^{\text{ap}}(x, \tau) = \bar{a}(\tau)[\nabla \log p_{\sigma(\tau)}(x) + \nabla \ell_\tau(x)]$. Girsanov's theorem [4] gives, for the drift difference $\bar{a}\nabla\ell_\tau$ shared diffusion coefficient $\sqrt{\bar{a}}$,

$$\frac{dP^{\text{pr}}}{dP^{\text{ap}}} = \exp\left(-\int_0^T \sqrt{\bar{a}}\, \nabla \ell_\tau^\top\, dW^{\text{ap}}_\tau - \tfrac{1}{2}\int_0^T \bar{a}\, \|\nabla \ell_\tau\|^2\, d\tau\right). \tag{4.2}$$

Substituting the Itô identity (4.1) to eliminate the stochastic integral, exactly as in the main derivation, gives

$$\frac{dP^{\text{pr}}}{dP^{\text{ap}}} = e^{\ell_0(Z_0) - \ell_T(Z_T)} \exp\left(\int_0^T V_\tau(Z_\tau)\, d\tau\right), \tag{4.3}$$

$$V_\tau(x) := \partial_\tau \ell_\tau(x) + \bar{a}(\tau)\, \nabla \log p_{\sigma(\tau)}(x) \cdot \nabla \ell_\tau(x) + \tfrac{1}{2}\bar{a}(\tau)\|\nabla \ell_\tau(x)\|^2 + \tfrac{1}{2}\bar{a}(\tau)\Delta \ell_\tau(x). \tag{4.4}$$

Written directly in terms of the true score, (4.4) has no sign ambiguity; it is the drift-substituted, simplified form of V_τ used throughout, and it is what is actually implemented (with $\nabla \log p_{\sigma(\tau)} = -s_\theta$ if one prefers the EDM-convention s_θ, cf. Remark 3.1). Since $\sigma(t) = t$ and $\tau = T - t$, $\partial_\tau \ell_\tau(x) = -\partial_\sigma \ell(x, \sigma)|_{\sigma = \sigma(\tau)}$.

Discretization. Rather than a naive left-endpoint quadrature of the whole of (4.4), we use the exact, frozen-state telescoping identity for the $\partial_\tau \ell_\tau$ term, $\int_{\tau_{k-1}}^{\tau_k} \partial_\tau \ell_\tau(z_{k-1})\,d\tau = \ell_{\tau_k}(z_{k-1}) - \ell_{\tau_{k-1}}(z_{k-1})$, and a quadrature only for the remaining term $H(x,\sigma) := \bar{a}(\sigma)\nabla \log p_\sigma(x) \cdot \nabla \ell(x,\sigma) + \frac{1}{2}\bar{a}(\sigma)\|\nabla \ell(x,\sigma)\|^2 + \frac{1}{2}\bar{a}(\sigma)\Delta \ell(x,\sigma)$. The incremental log-weight for a step from σ_{cur} to σ_{next} at the frozen pre-step state z is

$$\log \hat{G} = \big[\ell(z;\sigma_{\text{next}}) - \ell(z;\sigma_{\text{cur}})\big] + \int_{\sigma_{\text{next}}}^{\sigma_{\text{cur}}} H(z,\sigma)\,d\sigma, \tag{4.5}$$

with a one-time correction term $\ell(z_0;\sigma_{\max})$ added before the first step, reproducing the $e^{\ell_0(Z_0)}$ boundary factor in (4.3) (the discrete analogue of absorbing the initial condition into the first incremental weight). We combine (4.5) with adaptive systematic resampling, triggered whenever the effective sample size drops below half the particle count.

5 Numerical results

We fix $y = 0.5$, $r = 1$, giving a genuinely bimodal posterior with weights $(0.182, 0.818)$ on the two components – an asymmetric but non-degenerate target. Particles are propagated on the rho-spaced σ-schedule of [3], $\sigma_{\min} = 0.02$, $\sigma_{\max} = 20$, with 1000 steps.

Histogram comparison. Figure 1 compares, at $N = 200{,}000$ particles: the exact-guided diffusion (using (2.5)), the uncorrected approximate-guided diffusion (using (2.6) with no reweighting), the SMC-corrected approximate-guided diffusion (reweighted by (4.5) and resampled), and the analytical posterior density. The exact-guided diffusion matches the analytical posterior to within sampling noise (maximum absolute density deviation 0.004, consistent with the Poisson floor at this particle count), confirming that the base SDE machinery – score, discretization, drift sign – is correct independently of any approximation. The uncorrected approximate-guided diffusion is visibly biased: narrower, mode-collapsed toward a single peak (maximum deviation 0.294, about 67× larger). The SMC-corrected diffusion recovers the bimodal shape closely, though not perfectly (maximum deviation in the 0.01–0.07 range depending on run and bin width) – the residual gap is discussed below.

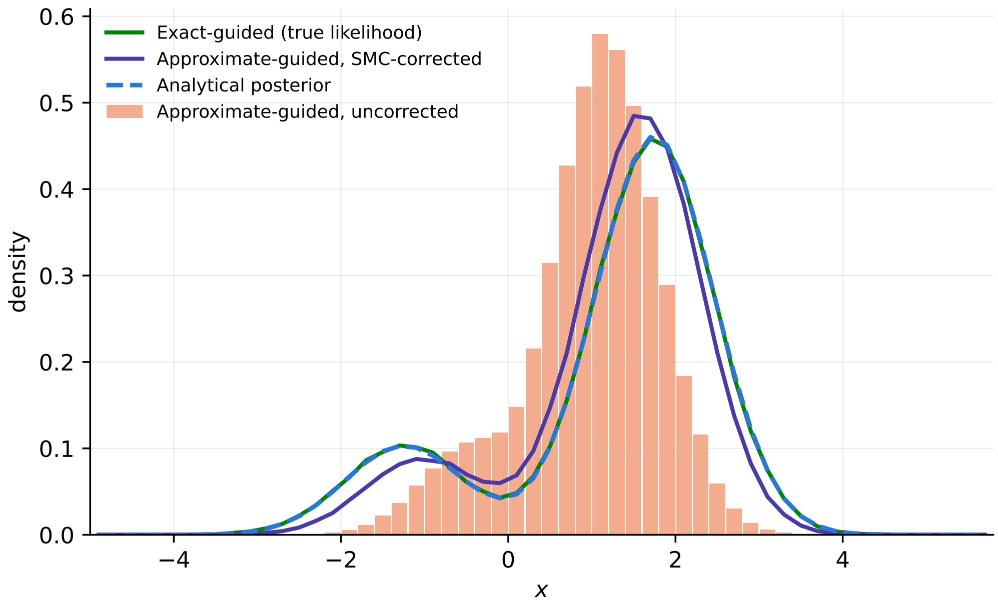

Figure 1: Density comparison at $N = 200{,}000$ particles. The exact-guided diffusion and the SMC-corrected approximate-guided diffusion both recover the bimodal analytical posterior; the uncorrected approximate-guided diffusion collapses toward a single mode.

Convergence in the number of particles. We estimated the posterior mean $\mathbb{E}[x \mid y]$ from the SMC-corrected sampler at $N \in \{10^3, 4 \times 10^3, 1.6 \times 10^4, 6.4 \times 10^4, 2 \times 10^5\}$, with 4–10 independent replications per N (fewer at the largest N, for compute reasons). Figure 2, left panel, shows both the RMSE against the true posterior mean (1.2027) and the standard deviation across replications, on a log-log scale, against the canonical $1/\sqrt{N}$ reference. A least-squares fit of log(std) against $\log N$ gives slope -0.46, close to the theoretical -0.5, confirming the estimator converges at the usual Monte Carlo rate. The right panel shows the raw bias (estimate minus truth) with ± 1 standard-error bars from the finite replication count; at every particle count but the largest, the error bar crosses zero, so the data cannot statistically distinguish a genuine residual bias from replication noise at that N. A parametric fit $\mathrm{MSE}(N) \approx b^2 + C/N$ under a non-negativity constraint on b^2 collapses to $b^2 = 0$, $C \approx 93$ – i.e., the data as collected is fully consistent with a purely canonical C/N decay and no detectable bias floor, though this should not be read as disproving a small bias: a separate experiment varying the number of discretization *steps* at fixed N found the histogram deviation improving sharply from 250 to 500 steps but then plateauing from 500 to 2000 steps, consistent with a small residual bias tied to the time-discretization order rather than to particle count. Distinguishing these possibilities conclusively would require substantially more replications concentrated at large N, where the current error bars are still wide relative to the effect size in question.

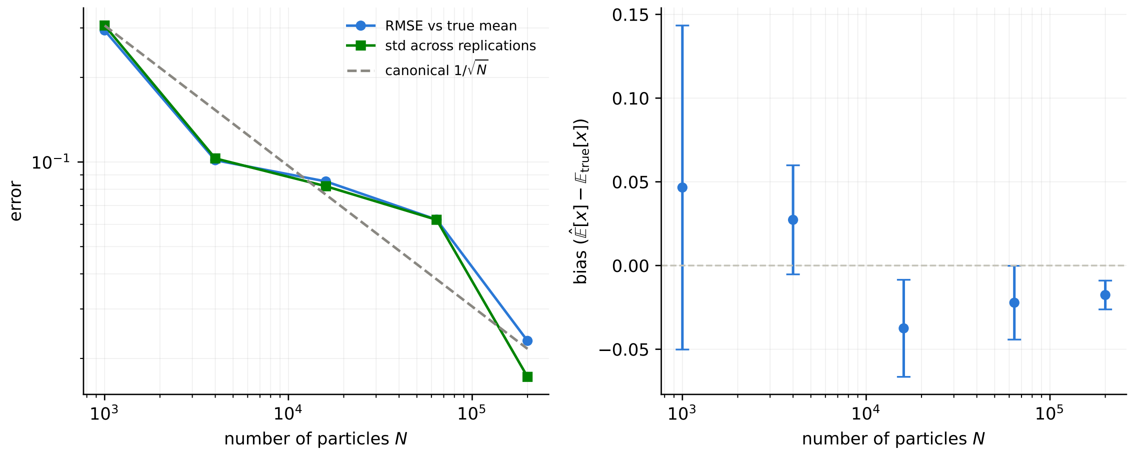

Figure 2: Left: RMSE and across-replication standard deviation of the SMC-corrected posterior-mean estimate versus particle count, log-log scale, against the canonical $1/\sqrt{N}$ reference. Right: bias (estimate minus true posterior mean) with ± 1 standard-error bars from the replication count.

The same convergence is visible directly in the sampled density itself. Figure 3 overlays the SMC-corrected histogram (a single representative run, not averaged over replications) against the analytical posterior at $N = 1{,}000$, 16,000, and 200,000 particles. At $N = 1{,}000$ the histogram is visibly jagged, though the bimodal shape is already present; at $N = 16{,}000$ it is smoother but still wobbles around the analytical curve, with a maximum deviation (0.036) that happens to read slightly *worse* than the $N = 1{,}000$ run (0.023) on this single-run metric alone – consistent with the large run-to-run variance at low N already seen in Figure 2, not a reversal of the trend. At $N = 200{,}000$ the histogram sits almost exactly on the analytical curve (maximum deviation 0.010). Read together with Figure 2, the message is the same one stated quantitatively above: the single-run histogram error shrinks with N on average, but any individual run at moderate N can be noisier than a smaller-N run purely by chance, which is exactly why the particle-count sweep in Figure 2 was run with multiple replications rather than read off a single realization.

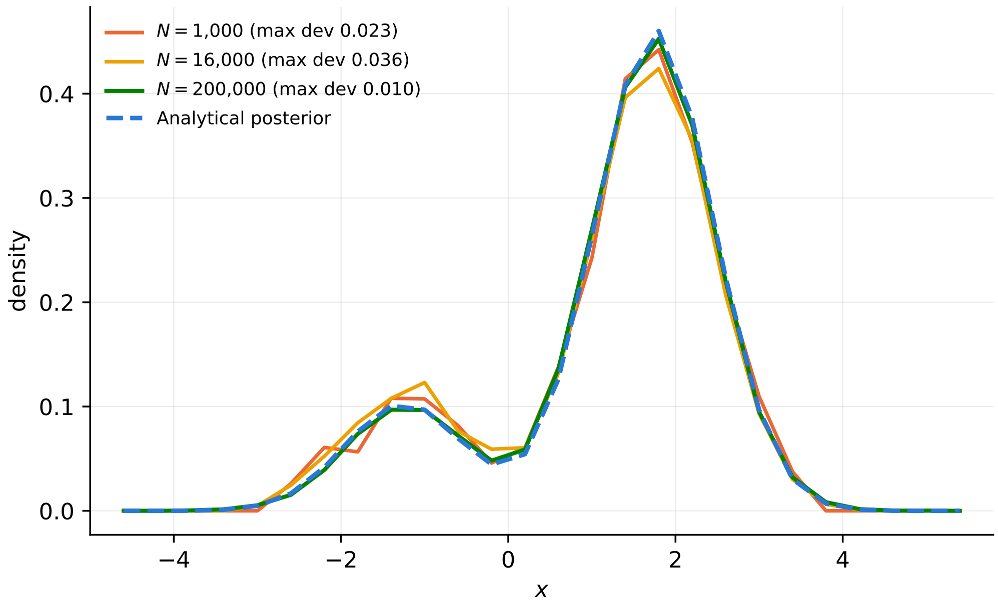

Figure 3: SMC-corrected density histogram (single representative run) at three particle counts, against the analytical posterior. Each run uses the same 1000-step discretization and the same adaptive resampling scheme.

6 Discussion

This toy model isolates the Girsanov/Doob-transform SMC correction from network-approximation error entirely, since the exact score, denoiser, posterior, and intermediate likelihood are all closed form. It served two purposes here: it exposed a genuine sign-convention error in the drift of the reverse-time SDE (Remark 3.1) before that error could propagate silently into results on the real PDE problem, and – once corrected – it gave a positive, closed-loop confirmation that the corrected SMC weight (4.5) converges to the true posterior at the canonical Monte Carlo rate in the particle count, up to a small residual tied to the time-discretization step count rather than to the correction formula itself. A natural extension, left for future work, is a moderate-dimensional ($d \sim 10$–20) version of the same construction, still fully closed form, in which the Laplacian term is a genuine trace over many dimensions and the Hutchinson estimator used on the real PDE problem can be checked directly against an exact analytical trace.

References

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